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**Hunters Hill**  
**High School**

**2020** Trial Examination

# Mathematics Extension 1

**General  
Instructions**

Reading time – 10 minutes  
Working time – 2 hours  
Write using black pen  
Calculators approved by NESA may be used  
A reference sheet is provided at the back of this paper  
In Questions 11-14, show relevant mathematical reasoning and/or calculations

**Total Marks:  
70**

**Section I – 10 marks** (pages 3-6)  
Attempt all Questions 1-10  
Allow about 15 minutes for this section

**Section II – 60 marks** (pages 7-12)  
Attempt Questions 11-14  
Allow about 1 hour and 45 minutes for this section

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**Section I****10 marks****Attempt Questions 1–10****Allow about 15 minutes for this section**Use the multiple-choice answer sheet for Question 1–10.

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1. In how many way can the letters of the word SUCCESS be arranged?

(A)  $7!$

(B)  $\frac{7!}{2!}$

(C)  $\frac{7!}{2! 3!}$

(D)  $4!$

2. The parametric equations  $x = ct, y = \frac{c}{t}$  form a

(A) Line

(B) Parabola

(C) Circle

(D) Rectangular Hyperbola

3. Which if the following vectors is perpendicular to  $\begin{bmatrix} -2 \\ 3 \end{bmatrix}$ ?

(A)  $\begin{bmatrix} -2 \\ 3 \end{bmatrix}$

(B)  $\begin{bmatrix} -3 \\ 2 \end{bmatrix}$

(C)  $\begin{bmatrix} -6 \\ -4 \end{bmatrix}$

(D)  $\begin{bmatrix} -4 \\ 6 \end{bmatrix}$

4. The parametric form of a function is given as  $y = \sin t + 1$ ,  $x = \cos t$ .  
The Cartesian form of the function is

(A)  $y = x + 1$

(B)  $x^2 = y^2$

(C)  $x^2 + (y - 1)^2 = 1$

(D)  $\tan t = \frac{y - 1}{x}$

5. The quadratic equation  $3x^2 - 5x - 6 = 0$  has roots of  $\alpha$  and  $\beta$ .  
The value for the expression  $\alpha^2 + \beta^2$  is

(A)  $\frac{5}{3}$

(B)  $\frac{61}{9}$

(C)  $\frac{-11}{9}$

(D)  $-2$

6. Which of the following expressions is equivalent to  $2 \sin 5\theta \cos 3\theta$

(A)  $\sin 8\theta + \sin 2\theta$

(B)  $\cos 8\theta + \cos 2\theta$

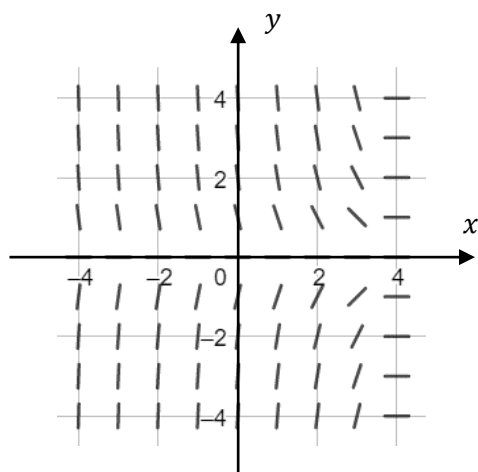
(C)  $\sin 8\theta - \sin 2\theta$

(D)  $\cos 8\theta - \cos 2\theta$

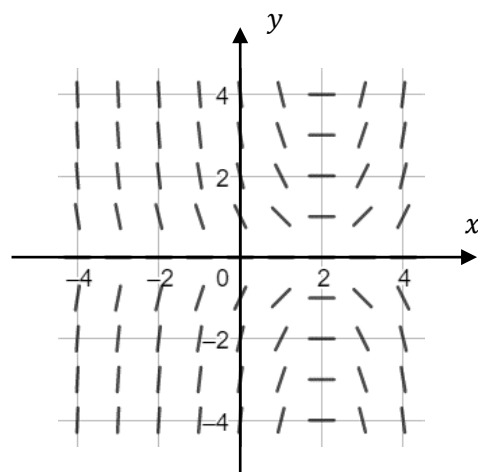
7. A differential equation is given by  $y' = xy - 2y$

Which of the following slope fields best represents the differential equation?

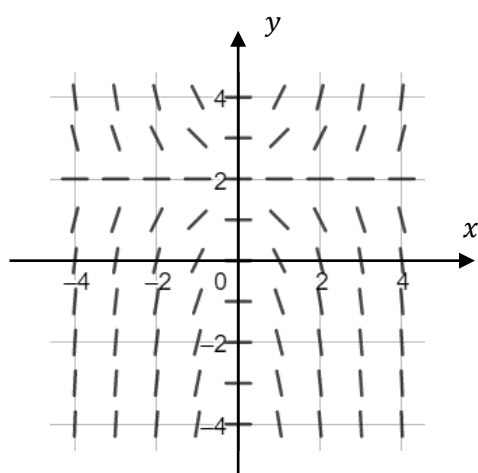
(A)



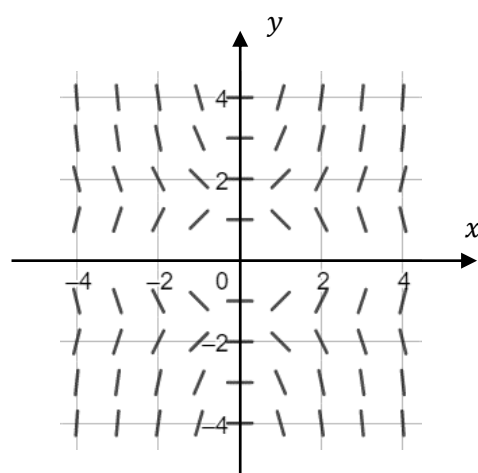
(B)



(C)



(D)



8. If  $A$  and  $B$  are the points  $(-1, 4)$  and  $(3, 7)$  respectively, then the unit vector in the direction of  $\overrightarrow{AB}$  is

(A)  $\frac{2}{5}\mathbf{i} + \frac{11}{5}\mathbf{j}$

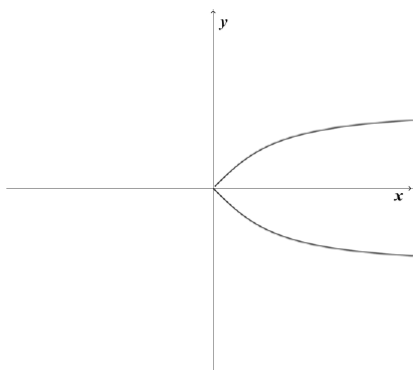
(B)  $4\mathbf{i} + 3\mathbf{j}$

(C)  $-3\mathbf{i} + 28\mathbf{j}$

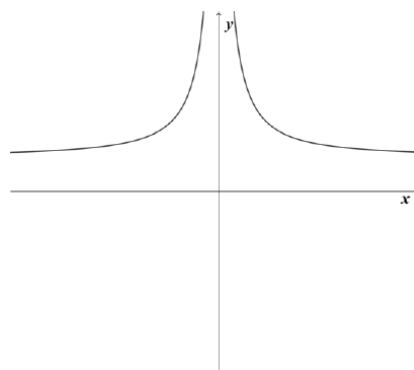
(D)  $\frac{4}{5}\mathbf{i} + \frac{3}{5}\mathbf{j}$

9. Which of the following sketches best represents  $|y| = \tan^{-1}(x)$ ?

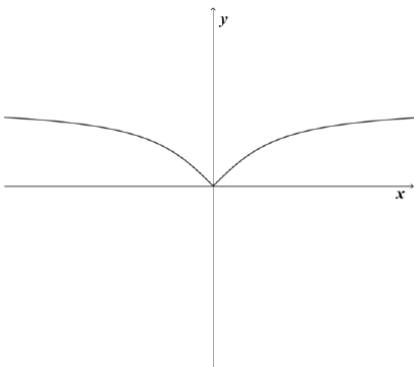
(A)



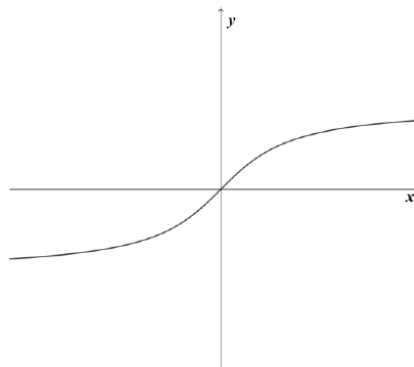
(B)



(C)



(D)



10. The letters A, B, C and D are used to form a four letter word.  
How many words can be written such that D comes before A?

- (A) 3
- (B) 6
- (C) 12
- (D) 24

End of Section I

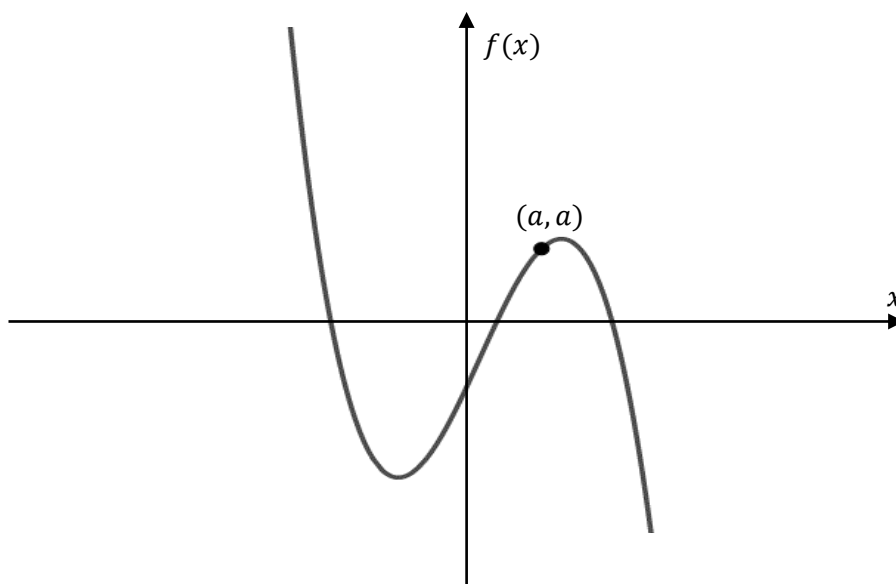
**Section II****60 marks****Attempt Questions 11–14****Allow about 1 hours and 45 minutes for this section**

Answer each question in a separate writing booklet.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

**Question 11** (15 marks) Begin a new Writing Booklet

- (a) Solve  $\frac{x}{x-1} < 3$ . 2
- (b) Solve  $|2x + 3| \leq 7$ . 1
- (c) The sketch below shows the curve  $y = f(x)$ , with the point  $(a, a)$  shown. 2  
Copy or trace this into your writing booklet and sketch the inverse relation,  $y = f^{-1}(x)$  on the same axes.

**Question 11 continues on next page**

(d) The cubic equation  $x^3 + 3Ax^2 - 4A = 0$ , where  $A > 0$ , has roots  $\alpha, \beta$  and  $\alpha + \beta$ .

(i) Use the sum of roots to show that  $\alpha + \beta = -\frac{3}{2}A$ . 1

(ii) Use the sum of the products of pairs of roots to show that  $\alpha\beta = -\frac{9}{4}A^2$ . 2

(iii) Show that  $A = \frac{2}{\sqrt{3}}$ . 2

(e) Find

(i)  $\int \frac{-1}{\sqrt{\frac{1}{4} - x^2}} dx$  2

(ii)  $\int \sin^2 2x dx$  3

End of Question 11

**Question 12** (15 marks) Begin a new Writing Booklet

- (a) Using the substitution  $u = \sin x$ , evaluate 3

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{3 \cos x}{1 + \sin x} dx$$

- (b) An extended family sits down to dinner at a round table. The family consists of 5 adults, and 2 children.

Find the number of arrangements possible

- (i) If there are no restrictions placed on seating. 1
- (ii) If two of the adults must sit together. 1
- (iii) If neither of the children can be seated together. 2

- (c) Consider the equation  $\cos x - 2 \sin x = 1$ , for  $-\pi \leq x \leq \pi$ .

- (i) Show that the equation can be written as  $t^2 + 2t = 0$ , where  $t = \tan \frac{x}{2}$ . 2
- (ii) Hence, solve the equation for  $-\pi \leq x \leq \pi$ , giving solutions to the 2 decimal places where necessary. 2

- (d) Water is being drained from a spout in the bottom of a cylindrical tank. According to Torricelli's Law, the volume  $V$  of water left in the tank obeys the differential equation

$$\frac{dV}{dt} = -k\sqrt{V}$$

where  $k$  is a constant.

- (i) Use separation of variables to find the general solution to this equation. 1
- (ii) Suppose the tank initially holds 30L of water, which initially drains at a rate of 1.8 L/min. How long will it take for the tank to drain completely? 3

**End of Question 12**

**Question 13** (15 marks) Begin a new Writing Booklet

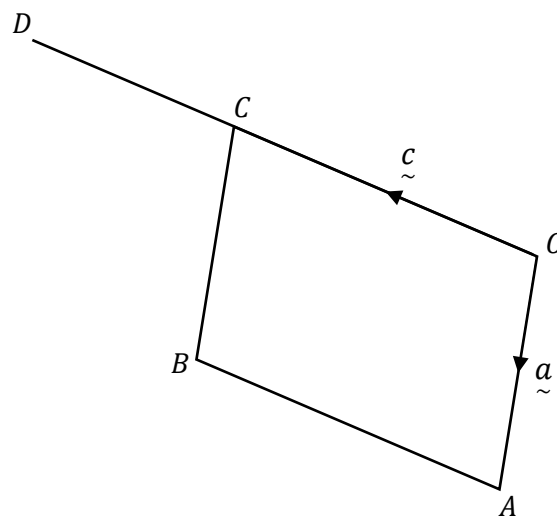
- (a) The surface area of a sphere ( $S = 4\pi r^2$ ) of radius  $r$  metres is decreasing at a rate of  $0.8 \text{ m}^2/\text{s}$  at an instant when  $r = 2.3$ .  
Calculate the rate of decrease, at this instant, of the radius of the sphere. 2

- (b) Prove by mathematical induction,  $a + ar + ar^2 + \dots + ar^{n-1} = \frac{a(r^n - 1)}{r - 1}$ ,  
for all integers  $n \geq 1$ . 3

- (c) (i) Prove the trigonometric identity  $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ . 3

- (ii) Hence, find expressions for the exact values of the solutions to the  
equation  $8x^3 - 6x = 1$ . 3

- (d) In the parallelogram  $OABC$ ,  $\overrightarrow{OA} = \underline{\underline{a}}$  and  $\overrightarrow{OC} = \underline{\underline{c}}$ .



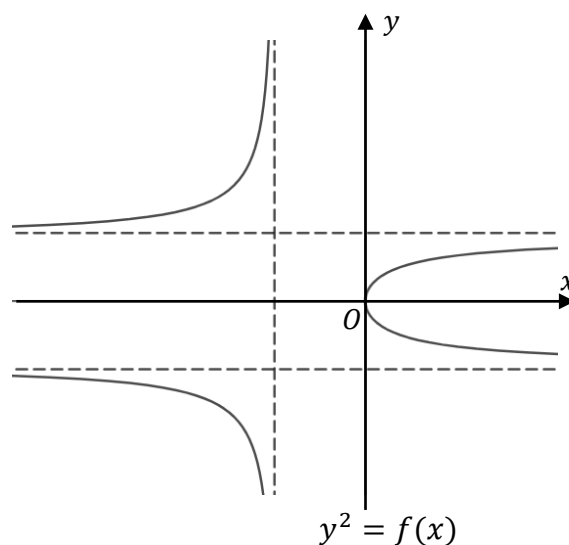
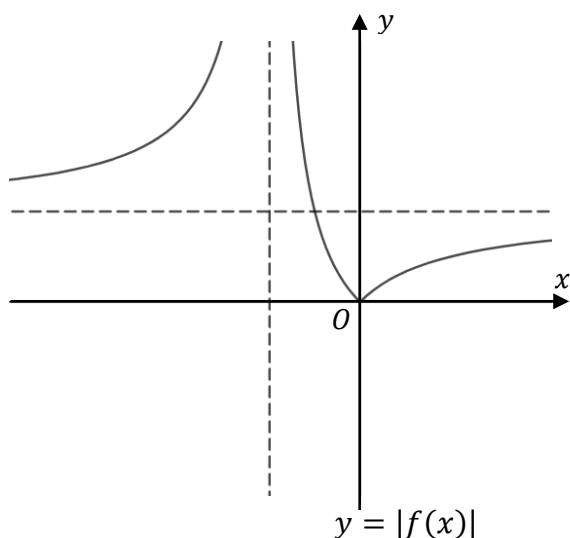
$X$  is the midpoint of the line  $AC$ , and  $OCD$  is a straight line such that the ratio  $OC : CD = k : 1$ .

- (i) Find  $\overrightarrow{OX}$  in terms of  $\underline{\underline{a}}$  and  $\underline{\underline{c}}$ . 1
- (ii) Write an expression for  $\overrightarrow{OD}$ . 1
- (iii) Given  $\overrightarrow{XD} = 3\underline{\underline{c}} - \frac{1}{2}\underline{\underline{a}}$ , find the value of  $k$ . 2

**End of Question 13**

**Question 14** (15 marks) Begin a new Writing Booklet

- (a) The graphs of  $y = |f(x)|$  and  $y^2 = f(x)$  are given below.



Draw separate one-third page sketches of the following curves, clearly indicating any important features such as turning points or asymptotes.

(i)  $y = f(x)$  1

(ii)  $y = f(|x|)$  1

- (b) Four friends go to a restaurant that serves 4 different main courses.

(i) In how many ways can the friends select meals from the courses available? 1

(ii) If each of the friends randomly chooses which meal they have, what is the probability that exactly two of the main course options are not chosen? 2

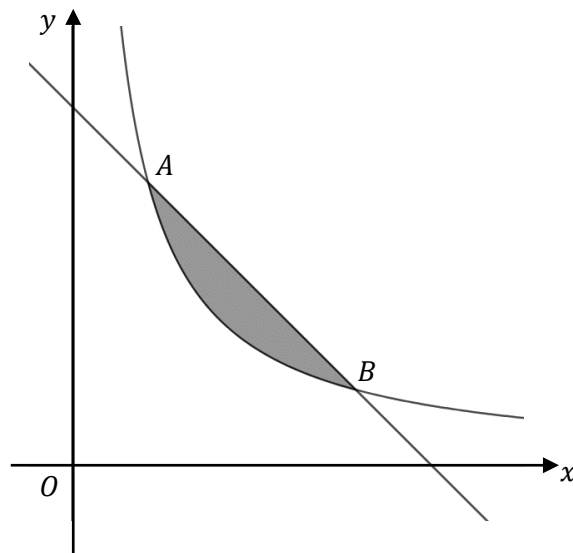
Question 14 continues on next page

- (c) A force described by the vector  $\mathbf{F} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$  newtons is applied to an object lying on the line  $l$  which is parallel to the vector  $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$

- (i) Find the component of the force  $\mathbf{F}$  in the direction of the line  $l$ . 3
- (ii) What is the component of the force perpendicular to the line  $l$ ? 1
- (iii) What force parallel to the line  $l$  must be applied to the object in order for it to remain stationary? 1

- (d) The diagram below shows the graphs of  $y = \frac{2}{x}$  and  $y = 3 - x$  for  $x > 0$ .

The shaded area is enclosed between the two graphs and their points of intersection  $A$  and  $B$ , as shown.



- (i) Find the coordinates of the points  $A$  and  $B$ . 2
- (ii) The shaded area is rotated about the  $y$ -axis. Find the exact volume of the solid formed. 3

**End of Examination**

Mathematics Extension 1 Trial  
**Section I Answer Sheet**

Student Number

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**Completely fill** the response oval representing the most correct answer.

- |     |   |                       |   |                       |   |                       |   |                       |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 2.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 3.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 4.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 5.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 6.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 7.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 8.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 9.  | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |
| 10. | A | <input type="radio"/> | B | <input type="radio"/> | C | <input type="radio"/> | D | <input type="radio"/> |

# 2020 12MAX TRIAL - H H H S - SOLUTIONS

1. C. 7 letters 35, 2C

$$\frac{7!}{3!2!}$$

2. D  $y = \frac{c}{x^2} = \frac{c^2}{x}$

3. C if  $\begin{bmatrix} a \\ b \end{bmatrix}$  is perpendicular,  
then  $-2a + 3b = 0$

$$\frac{a}{b} = \frac{3}{2}$$

$\begin{bmatrix} -6 \\ -4 \end{bmatrix}$  satisfies this

4. C  $\sin^2 t = (y-1)^2$ ,  $\cos^2 t = x$ .  
 $\therefore x^2 + (y-1)^2 = 1$

5. B  $\alpha + \beta = \frac{5}{3}$   $\alpha\beta = 2$ .

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= \frac{25}{9} + 4$$

$$= \frac{61}{9}$$

7. B

$$y' = y(x-2) \quad x=2 \Rightarrow y'=2$$

8. D

$$\vec{AB} = (3, 7) - (-1, 4)$$

$$= \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$|\vec{AB}| = \sqrt{4^2 + 3^2}$$

$$= 5$$

9. A

10. C

$$\begin{array}{r} \text{D} \\ \hline \text{D} \\ \hline \text{D} \end{array} \quad \begin{array}{r} 3 \times 2! \\ 2 \times 2! \\ 2! \\ \hline 12 \end{array}$$

11. a

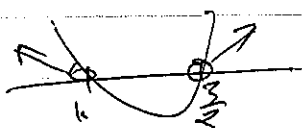
$$\frac{x}{x-1} \leq 3$$

$$x(x-1) \leq 3(x-1)^2$$

$$3(x-1)^2 - x(x-1) \geq 0$$

$$(x-1)(3x-3-x) \geq 0$$

$$(x-1)(2x-3) \geq 0$$



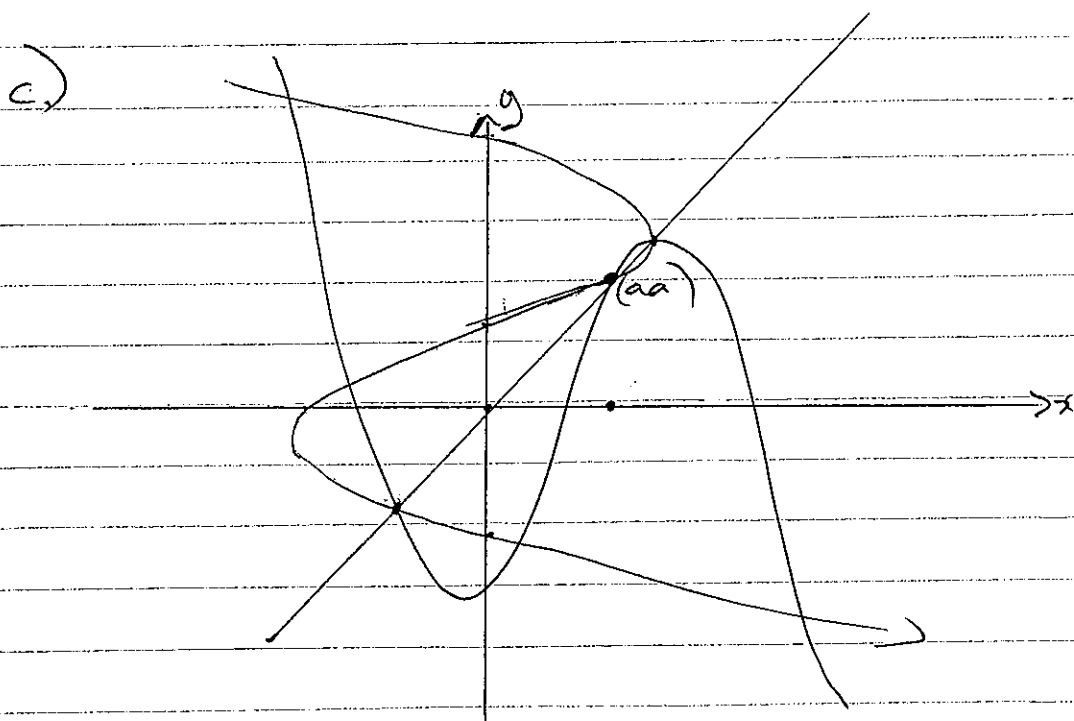
$$\therefore x < 1, x > \frac{3}{2}$$

b)  $|2x+3| \leq 7$

$$-7 \leq 2x+3 \leq 7$$

$$-10 \leq 2x \leq 4$$

$$-5 \leq x \leq 2$$



d) i)  $x^3 + 3Ax^2 - 4A = 0$ ,  $\alpha, \alpha+\beta, \beta$

$$\alpha + \beta + (\alpha + \beta) = \frac{-3A}{1}$$

$$\therefore 2(\alpha + \beta) = -3A$$

$$\alpha + \beta = \frac{-3A}{2}$$

ii)  $\alpha\beta + \alpha(\alpha+\beta) + \beta(\alpha+\beta) = \frac{0}{1}$

$$\alpha\beta + (\alpha+\beta)^2 = 0$$

$$\alpha\beta = -(\alpha+\beta)^2$$

$$= -\frac{9A^2}{4}$$

$$iii) \quad \alpha\beta(\alpha+\beta) = -4A$$

$$-\frac{9}{4}A^2 \times \left(-\frac{3}{2}A\right) = -4A$$

$$-\frac{27}{8}A^3 = -4A$$

$$-27A^2 = -36, \quad A > 0$$

$$A^2 = \frac{36}{27} \quad \frac{32}{27}$$

$$\therefore A = \frac{6}{3\sqrt{3}} \quad \frac{4\sqrt{3}}{3\sqrt{3}}$$

$$= \frac{2}{\sqrt{3}}$$

$$e) i) \quad \int \frac{-1}{\sqrt{\frac{1}{4}-x^2}} dx = \cos^{-1} 2x + C$$

$$ii) \quad \int \sin^2 2x dx = \frac{1}{2} \int (1 - \cos 4x) dx \quad \cos 2x = 1 - 2\sin^2 x$$

$$= \frac{1}{2} \left( x - \frac{\sin 4x}{4} \right) + C$$

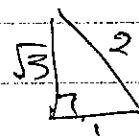
Q12.

a) 
$$I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{3 \cos x}{1 + \sin x} dx$$

let  $u = \sin x$   $du = \cos x dx$ .

at  $x = \frac{\pi}{3}$ ,  $u = \frac{\sqrt{3}}{2}$

$x = \frac{\pi}{6}$ ,  $u = \frac{1}{2}$



$$I = 3 \int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \frac{du}{1+u}$$

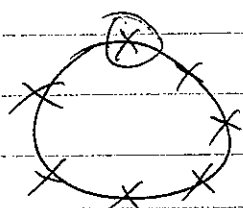
$$= 3 \left[ \ln |1+u| \right]_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}}$$

$$= 3 \left( \ln \left( 1 + \frac{\sqrt{3}}{2} \right) - \ln \left( 1 + \frac{1}{2} \right) \right)$$

$$= 3 \ln \left( \frac{2+\sqrt{3}}{3} \right)$$

$$= 0.6550368$$

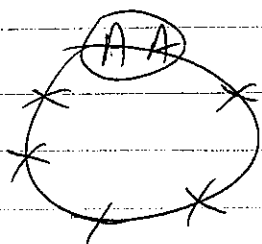
b) i)



7 people at table.

fix one  $\Rightarrow$  arrangements  $= 6!$   
 $= 720$

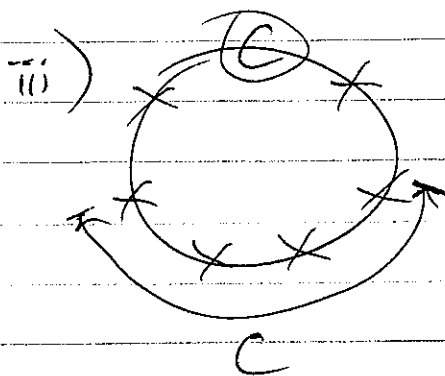
ii)



6 groups  $\Rightarrow 5!$

2 adults  $\Rightarrow 2!$

arrangements  $= 5! \cdot 2!$   
 $= 240$



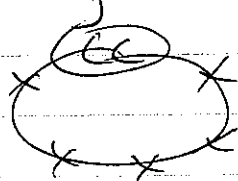
fix one child.

other child sits in one of four spots

remaining  $\rightarrow 5!$

$$\therefore \text{arrangements} = 5! \times 4 = 480$$

alternatively



$$\text{arrangements} = 720 - 5! \cdot 2! = 480$$

2) i)  $\cos x - 2 \sin x = 1, -\pi \leq x \leq \pi$

$$\cos x = \frac{1-t^2}{1+t^2}, \quad \sin x = \frac{2t}{1+t^2}$$

$$\therefore \frac{1-t^2}{1+t^2} - 2 \cdot \frac{2t}{1+t^2} = 1$$

$$1-t^2-4t = 1+t^2$$

$$\therefore 2t^2+4t=0$$

$$t^2+2t=0$$

ii)  $t(t+2)=0$

$$\therefore t=0, -2$$

$$\tan \frac{x}{2} = 0, -2$$

$$\therefore \frac{x}{2} = \tan^{-1} 0, \tan^{-1}(-2)$$

$$x = 0, 2 \tan^{-1}(-2)$$

$$= 0, -2.214297$$

d) i)  $\frac{dV}{dt} = -k\sqrt{V}$

$$\int \frac{dV}{\sqrt{V}} = \int -k dt$$

$$2V^{\frac{1}{2}} = -kt + c$$

$$V = \frac{1}{4}(c - kt)^2$$

ii)  $t=0, V=30, \frac{dV}{dt} = -1.8$

$$30 = \frac{1}{4}(c - k(0))^2$$

$$c = 2\sqrt{30}$$

$$-1.8 = -k\sqrt{30}$$

$$k = \frac{9}{5\sqrt{30}}$$

$$\therefore V = \frac{1}{4} \left( 2\sqrt{30} - \frac{9}{5\sqrt{30}} t \right)^2$$

for  $V=0$   $2\sqrt{30} - \frac{9}{5\sqrt{30}} t = 0$

$$t = 2\sqrt{30} \times \frac{5\sqrt{30}}{9}$$

$$= 33\frac{1}{8} \text{ min}$$

Q13

a)  $\frac{dS}{dt} = 0.8 \text{ m}^2/\text{s}$ ,  $S = 4\pi r^2$ , find  $\frac{dr}{dt}$

$$\frac{dS}{dr} = 8\pi r.$$

$$\begin{aligned} \frac{dr}{dt} &= \frac{dr}{dS} \cdot \frac{dS}{dt} \\ &= \frac{1}{8\pi r} \cdot 0.8 \end{aligned}$$

$$\begin{aligned} \text{at } r=2.3 \quad \frac{dr}{dt} &= \frac{0.8}{8\pi(2.3)} \\ &= 0.0138396 \text{ ms}^{-1} \end{aligned}$$

b)  $a + ar + ar^2 + \dots + ar^{n-1} = a \frac{(r^n - 1)}{r - 1}$ ,  $n \geq 1$

Prove true for  $n=1$

$$\text{LHS} = a$$

$$\text{RHS} = a \frac{(r-1)}{r-1}$$

$$= a = \text{LHS}$$

$\therefore$  true for  $n=1$

Assume true for  $n=k$

$$\therefore a + ar + ar^2 + \dots + ar^{k-1} = a \frac{(r^k - 1)}{r - 1}$$

induction hypothesis

Prove true for  $n=k+1$

$$\text{i.e. } a + ar + ar^2 + \dots + ar^{k-1} + ar^k = a \frac{(r^{k+1} - 1)}{r - 1}$$

$$\text{LHS} = a + ar + ar^2 + \dots + ar^{k-1} + ar^k$$

$$= a \frac{(r^k - 1)}{r - 1} + ar^k \quad \text{using hypothesis}$$

$$= a \left[ \frac{r^k - 1}{r - 1} + r^k \right]$$

$$= a \left[ \frac{r^k - 1 + r^k(r - 1)}{r - 1} \right]$$

$$= a \left[ \frac{r^k - 1 + r^{k+1} - r^k}{r - 1} \right]$$

$$= a \frac{(r^{k+1} - 1)}{r - 1}$$

$$= \text{RHS.}$$

$\therefore$  by principle of mathematical induction  
statement is true for all  $n \geq 1$ .

o) i)  $\cos 3\theta = \cos(2\theta + \theta)$

$$= \cos 2\theta \cos \theta - \sin 2\theta \sin \theta$$

$$= (2\cos^2 \theta - 1)\cos \theta - 2\sin \theta \cos \theta \sin \theta$$

$$= 2\cos^3 \theta - \cos \theta - 2\sin^2 \theta \cos \theta$$

$$= 2\cos^3 \theta - \cos \theta - 2(1 - \cos^2 \theta)\cos \theta$$

$$= 2\cos^3 \theta - \cos \theta - 2\cos \theta + 2\cos^3 \theta$$

$$= 4\cos^3 \theta - 3\cos \theta$$

ii)  $8x^3 - 6x = 1$

let  $x = \cos \theta$

$$\therefore 4\cos^3 \theta - 6\cos \theta = 1$$

$$2(4\cos^3 \theta - 3\cos \theta) = 1$$

$$\therefore 2\cos 3\theta = 1$$

$$\cos 3\theta = \frac{1}{2}$$

$$\begin{aligned} -\pi &\leq \theta \leq \pi \\ -3\pi &\leq 3\theta \leq 3\pi \end{aligned}$$



$$\cos 3\theta > 0 \quad 3\theta \text{ in } 1^{\text{st}} + 4^{\text{th}} \text{ quadrants}$$

$$\text{as } \cos \frac{\pi}{3} = \frac{1}{2}$$

$$\text{then } 3\theta = -2\pi - \frac{\pi}{3}, -2\pi + \frac{\pi}{3}, -\frac{\pi}{3}, \frac{\pi}{3}, 2\pi - \frac{\pi}{3}, 2\pi + \frac{\pi}{3}$$

$$\theta = -\frac{7\pi}{9}, -\frac{5\pi}{9}, -\frac{\pi}{9}, \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$$

$$x = \left(\cos\left(\frac{\pi}{9}\right), \cos\left(\frac{5\pi}{9}\right), \cos\left(\frac{7\pi}{9}\right)\right)$$

$\cos x$  is an even function

$$\begin{aligned} \text{d) i) } \vec{OX} &= \frac{1}{2}\underline{c} + \frac{1}{2}\underline{a} \\ &= \frac{1}{2}(\underline{a} + \underline{c}) \end{aligned}$$

$$\text{ii) } \vec{OD} = \frac{k+1}{k} \vec{OC} = \frac{k+1}{k} \underline{c}$$

$$\begin{aligned} \text{iii) } \vec{XD} &= \vec{OD} - \vec{OX} \\ &= \frac{k+1}{k} \underline{c} - \frac{1}{2}(\underline{a} + \underline{c}) \\ &= \left(\frac{k+1}{k} - \frac{1}{2}\right) \underline{c} - \frac{1}{2}\underline{a} \\ &= \frac{2k+2-k}{2k} \underline{c} - \frac{1}{2}\underline{a} \end{aligned}$$

$$= \frac{k+2}{2k} z - \frac{1}{2} a$$

$$\text{as } \frac{k+2}{2k} z - \frac{1}{2} a = 3z - \frac{1}{2} a$$

$$\frac{k+2}{2k} = 3 \quad (\text{equating } z \text{ component})$$

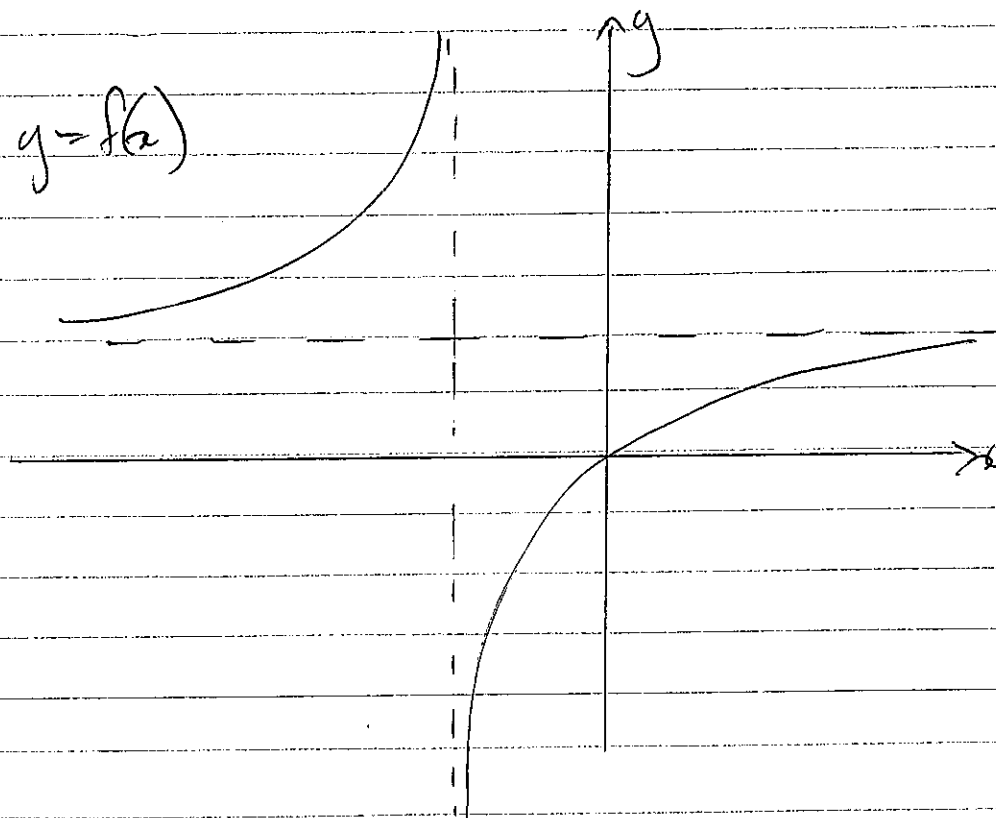
$$k+2 = 6k$$

$$5k = 2$$

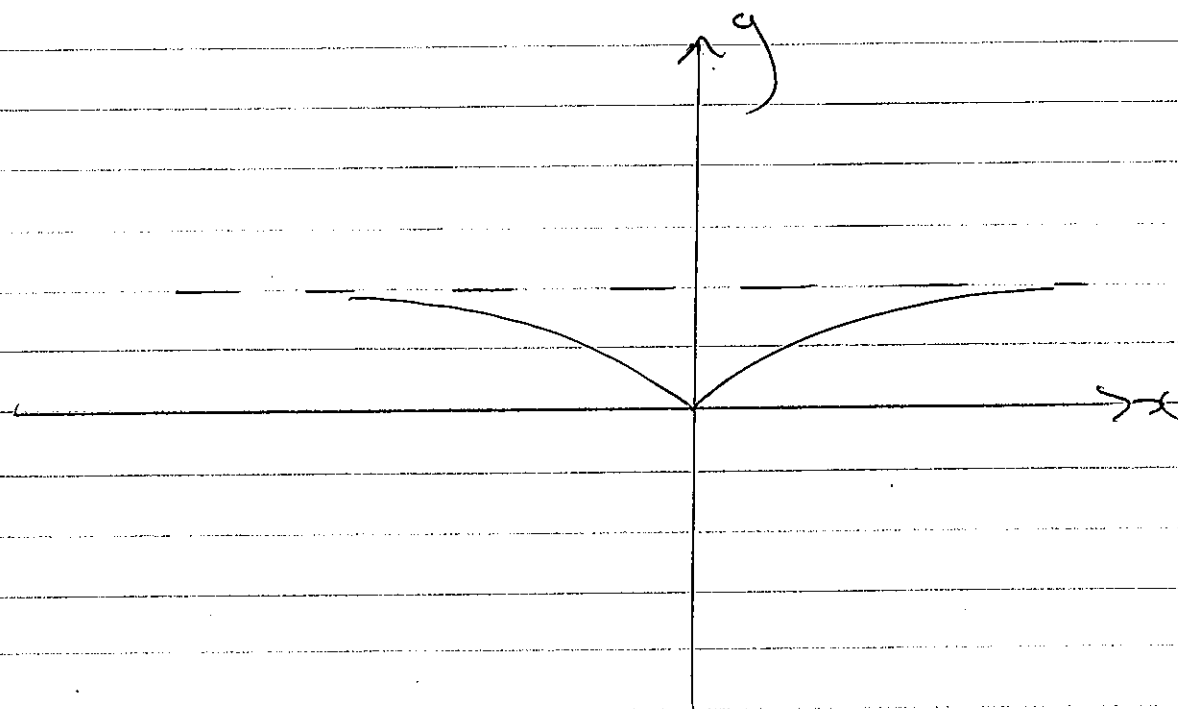
$$k = \frac{2}{5}$$

Q14

a) i)  $y = f(x)$



ii)



b) i) number of arrangements =  $4^4$   
 $= 256$

ii) ways that 4 meals can be had

3 1 / /

2 2 / /

for 3/1  ${}^4C_3 \times 2!$

for 2/2  ${}^4C_2 \times 2!$

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total for these two meals =  ${}^4C_3 \times 2! + {}^4C_2 \times 2!$   
 $= 20$

combinations of two meals =  ${}^4C_2$

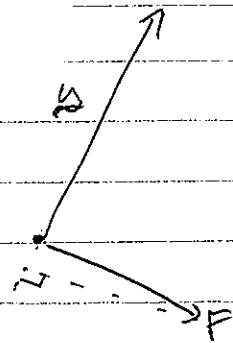
$$\text{Probability} = \frac{20 \times 4C_2}{256}$$

$$= \frac{5}{16}$$

c) i)  $F = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$

unit vector in direction of  $l$ .

$$\hat{u} = \frac{1}{\sqrt{29}} \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$



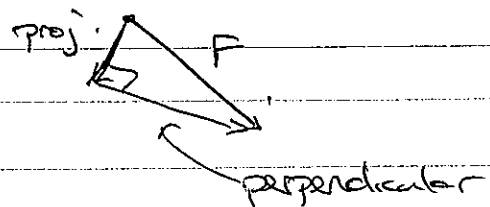
$$\text{proj}_{\hat{u}} F = (F \cdot \hat{u}) \hat{u}$$

$$= \left( 3 \times \frac{2}{\sqrt{29}} + -2 \times \frac{5}{\sqrt{29}} \right) \cdot \frac{1}{\sqrt{29}} \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$= \frac{-4}{29} \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

ii)

perpendicular component  
 $= F - \text{proj}_{\hat{u}} F$



$$= \begin{bmatrix} 3 \\ -2 \end{bmatrix} - \frac{4}{29} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \frac{1}{29} \begin{bmatrix} 85 \\ -58 \end{bmatrix}$$

ii) opposite of parallel component force applied to object. is

$$\frac{4}{29} \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

d) i)  $y = \frac{2}{x}$ ,  $y = 3 - x$ .

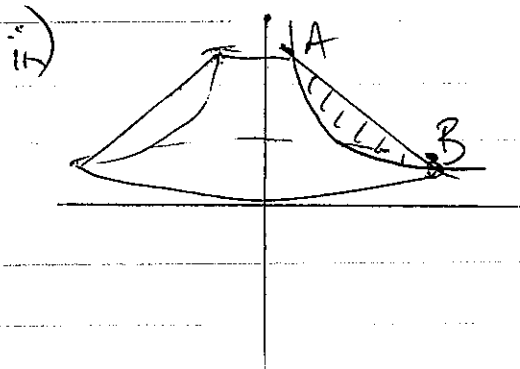
equating  $\frac{2}{x} = 3 - x$

$$x^2 - 3x + 2 = 0$$

$$(x-2)(x-1) = 0$$

$$\therefore x = 1, 2$$

$\therefore$  A is (1, 2) and B is (2, 1)



$$V = \pi \int_1^2 \left( (3-y)^2 - \left( \frac{2}{y} \right)^2 \right) dy$$
$$= \pi \int_1^2 \left( (3-y)^2 - \frac{4}{y^2} \right) dy$$

$$= \pi \left[ \frac{(3-y)^3}{-3} + \frac{4}{y} \right]_1^2$$
$$= \pi \left( \frac{1^3}{-3} + \frac{4}{2} - \left( \frac{2^3}{-3} + \frac{4}{1} \right) \right)$$
$$= \pi \left( 2 - \frac{1}{3} - 4 + \frac{8}{3} \right)$$
$$= \frac{\pi}{3} \text{ units}^3$$